

Fracture Mechanics Analysis of Semi-Infinite Plate

Fracture Mechanics

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Question-1

Given:

$$\sigma = 100\text{MPa}$$

$$b = 10\text{mm}$$

$$h = 40\text{mm}$$

$$a = 5\text{mm}$$

Objective:

- For a semi-infinite plate (plane stress condition) under mode I loading, simulate and find:
 - Stress intensity factor
 - Crack opening displacement δ
 - Displacement due to crack at remote points, Δ_{crack}
 - Crack opening area.
- Find the analytical solution and compare with the results from Abaqus.

Material Properties:

$$E = 207 \text{ GPa}$$

$$\nu = 0.3$$

Boundary conditions

- The top and bottom edges are given a pressure load of 100MPa as said in question.
- The plate is partitioned along the mid planes both in x- and y-direction. On the horizontal partition line, excluding the crack part and the line inside those circular partition, in remaining portion, Y-symmetry condition is applied ($U_2=U_{R1}=U_{R3}=0$).
- On the vertical partition line, X-symmetry condition is applied ($U_1=U_{R2}=U_{R3}=0$). (Fig 1.2)

Meshing Details

- Element type: CPS46
- A global seed element size of 1 is given and on the circular partition around cracktip, a local seed of 10 element numbers is given.

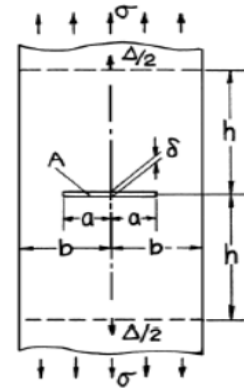


Fig 1.1: Mode I loading of semi-infinite plate (plane stress condition).

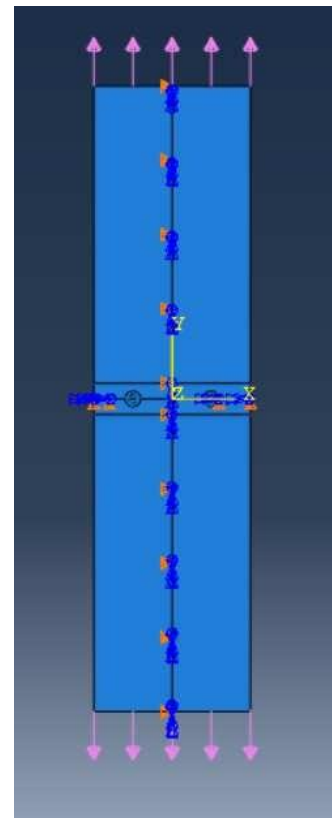


Fig 1.2: Loading and Boundary condition for Q1

- On mesh control, quad-dominated sweep mesh is applied inside the circular portioned portion, and in the remaining portion, quad-dominated free mesh is applied.

Fig 1.4 shows the final model after meshing, whereas Fig 1.3 gives a more clear picture of meshing near the crack tip.

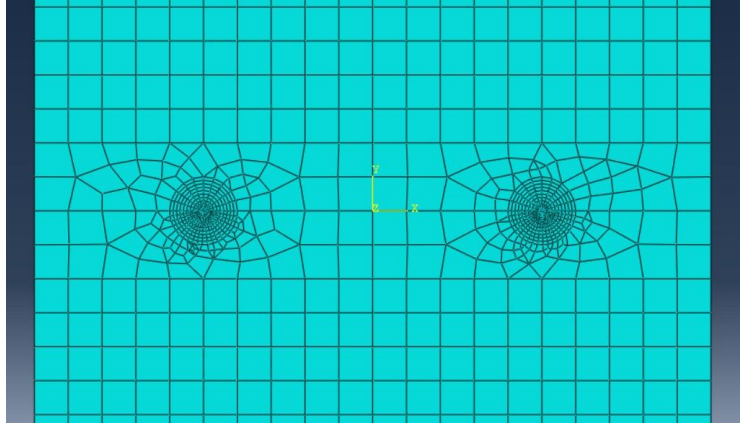


Fig 1.3: Mesh near the crack tip

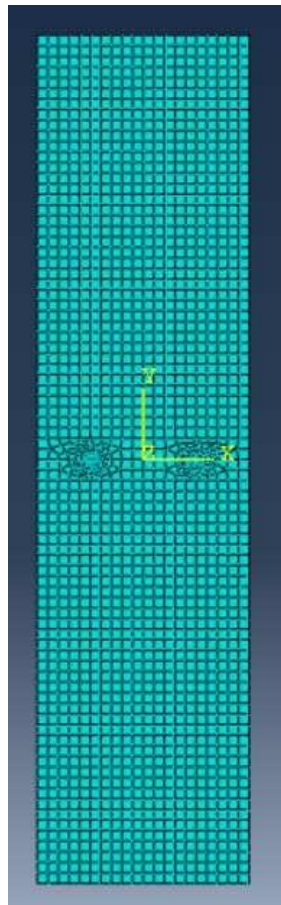


Fig 1.4: Meshed model of plate with center crack

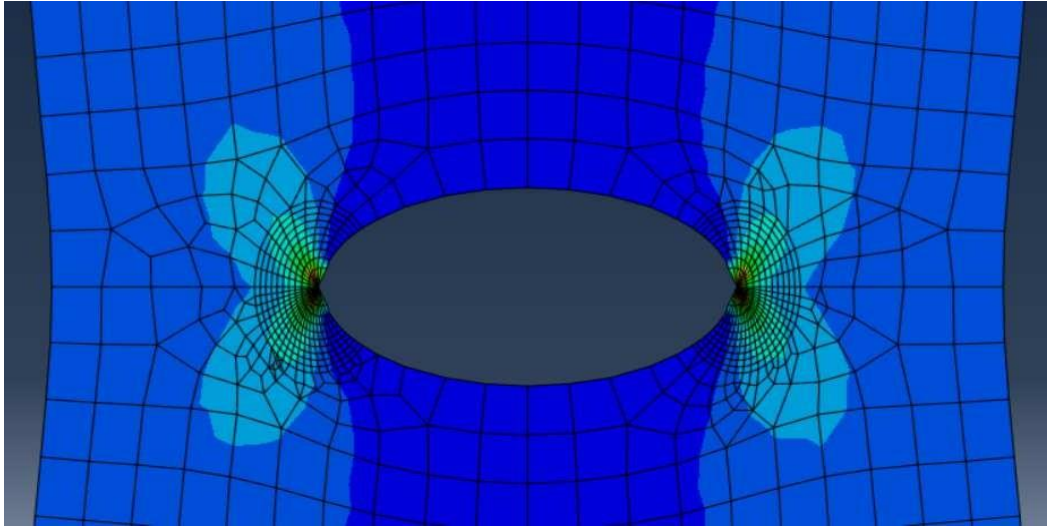


Fig 1.5: Stress contour near the crack

Results:

(a) Stress Intensity Factor

The stress intensity factor for mode I loading problem for semi infinite plate is given by,

$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

For given crack and width dimension, $F\left(\frac{a}{b}\right) = F(0.5) = 1.1867$ (Data from TADA file).

Therefore, $K_I = 470.33 \text{ MPa} \cdot \text{mm}^{\frac{1}{2}}$.

The Stress intensity factor obtained from Abaqus, $K_I = 468.049 \text{ MPa} \cdot \text{mm}^{\frac{1}{2}}$

Fig 1.5 shows the stress contour after applying the load.

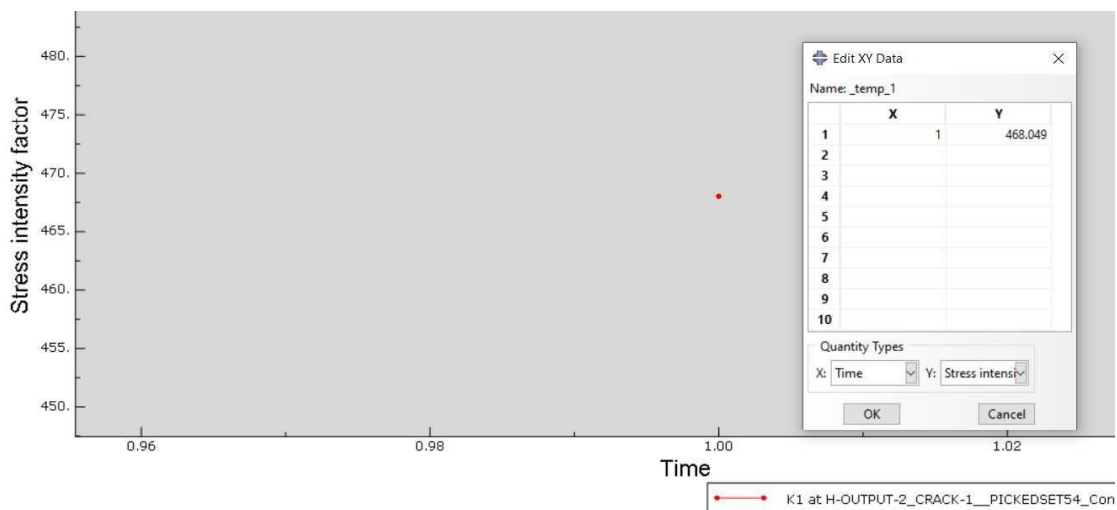


Fig 1.6: Stress Intensity factor obtained from Abaqus

(b) Crack opening displacement:

Scale:	3.71097e+002,	3.71097e+002,	3.71097e+002,	-
Deformed distance (unscaled):	-3.13246e-037,	1.14353e-002,	0.00000e+000,	1.14353e-002
Deformed distance (scaled):	-1.16245e-034,	4.24360e+000,	0.00000e+000,	4.24360e+000
Relative displacement (unscaled):	-3.13246e-037,	1.14353e-002,	0.00000e+000,	1.14353e-002

From Abaqus, the crack opening displacement at center is found to be, $\delta = 0.014353\text{mm}$.

Analytically, the crack opening displacement at center is given by,

$$\delta = \frac{4\sigma a}{E'} V_1(a/b)$$

where,

$$V_1\left(\frac{a}{b}\right) = -0.071 - 0.535\left(\frac{a}{b}\right) + 0.169\left(\frac{a}{b}\right)^2 - 0.090\left(\frac{a}{b}\right)^3 + 0.020\left(\frac{a}{b}\right)^4 - \frac{1.071}{\left(\frac{a}{b}\right)} \ln\left(1 - \frac{a}{b}\right)$$

For $\left(\frac{a}{b}\right) = 0.5$, $V_1\left(\frac{a}{b}\right) = 1.17847126$. Therefore, crack opening displacement at center is

$$\delta = 0.0113862\text{mm}$$

(c) Displacement due to crack at remote point:

Analytically, the displacement due to crack tip at remote point is given by,

$$\Delta_{crack} = \frac{4\sigma a}{E'} V_2(a/b)$$

where

$$V_2\left(\frac{a}{b}\right) = -1.071 + 0.250\left(\frac{a}{b}\right) - 0.357\left(\frac{a}{b}\right)^2 + 0.121\left(\frac{a}{b}\right)^3 - 0.047\left(\frac{a}{b}\right)^4 - 0.008\left(\frac{a}{b}\right)^5 - \frac{1.071}{\left(\frac{a}{b}\right)} \ln\left(1 - \frac{a}{b}\right)$$

For, $\left(\frac{a}{b}\right) = 0.5$, $V_2\left(\frac{a}{b}\right) = 0.4614$. Therefore

$$\Delta_{crack} = 4.458 \times 10^{-3}\text{mm}$$

From Abaqus, we get the original displacement at $h=40$

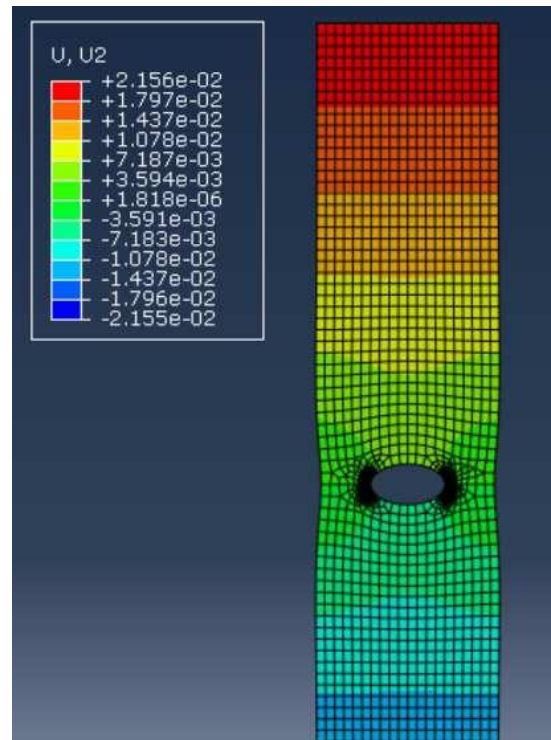


Fig 1.6: Nodal displacement contour in y-direction

$$\frac{\Delta_{total}}{2} = 2.156 \times 10^{-2} \text{ (See Fig 1.6).}$$

$$\Delta_{total} = 0.04312mm.$$

The displacement of node at h=40mm, when there is no crack is given by,

$$\Delta_{Nocrack} = \frac{\sigma}{E} \cdot 2h = \frac{100}{207000} \cdot 2 \times 40 = 0.038647mm$$

Therefore, from Abaqus, the displacement due to crack tip at remote point is given by

$$\Delta_{crack} = \Delta_{total} - \Delta_{Nocrack} = 0.04312 - 0.038647$$

$$\Delta_{crack} = 4.473 \times 10^{-3}mm.$$

(d) Crack Opening Area:

The crack opening area is given by,

$$A = \Delta_{crack} \cdot 2b$$

$$\text{From Analytical solution, } A = \Delta_{crack} \cdot 2b = 4.458 \times 10^{-3} \times 2 \times 10 = 0.08916mm^2$$

$$\text{From Abaqus, } A = \Delta_{crack} \cdot 2b = 4.473 \times 10^{-3} \times 2 \times 10 = 0.08946mm^2$$

Conclusion:

The following table shows the comparison of analytical result with Abaqus result

	stress intensity factor KI	crack opening displacement	Displacement due to crack tip at remote point	Crack opening area
Analytical Results	$470.33 MPa \cdot mm^{\frac{1}{2}}$	0.0113862mm	$4.458 \times 10^{-3}mm$	$0.08916mm^2$
Results from Abaqus	$468.049 MPa \cdot mm^{\frac{1}{2}}$	0.014353mm	$4.473 \times 10^{-3}mm$	$0.08946mm^2$

From the table, it is clear that the results from Abaqus are almost the same as the analytical solution.

Reference:

- The Stress analysis of Cracks Handbook, Hiroshi Tada, Paul C. Paris and George R. Irwin

Question-2

Given:

$$\sigma = 100\text{MPa}$$

$$b = 10\text{mm}$$

$$h = 40\text{mm}$$

Crack angle with respect to horizontal axis,
 $\beta=30^\circ$.

Since, the problem has to closely represent the centre crack in infinite plate, choosing $a/b=0.1$. That is, crack size = $2a = 2\text{mm}$.

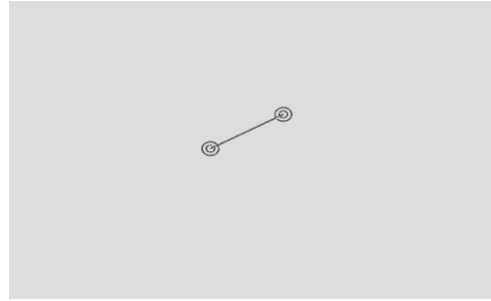


Fig 2.1: Crack inclined at angle 30° , created on Abaqus Model

Objective:

- For a semi-infinite plate (plane stress condition) with crack inclined at an angle mode $\beta=30^\circ$ with respect to horizontal axis, simulate and find:
 - Stress intensity factors K_I and K_{II}
 - Crack propagation angle
- Compare the results with the analytical solution

Material Property:

$$E = 207 \text{ GPa}$$

$$\nu = 0.3$$

Boundary conditions

- The plate is given a pressure of 100MPa on top and bottom edge. (Fig 2.2)

Meshing Details:

- Element Type: CPS4R
- Global seed of element size 1 is applied and on the circular partition around cracktip, a local seed of 5 element numbers is given.
- On mesh control, quad-dominated sweep mesh is applied inside the circular portioned portion, and in the remaining portion, quad-dominated free mesh is applied.

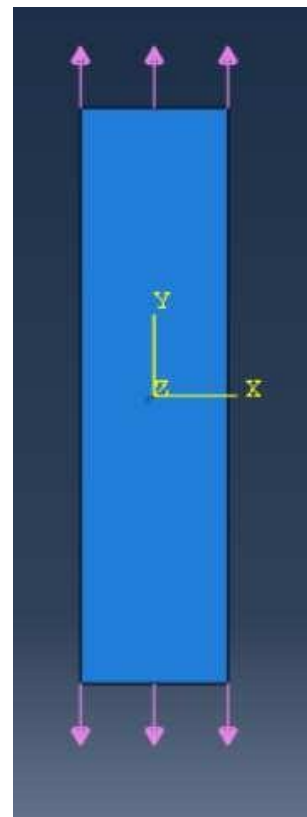


Fig 2.2: Loading Condition

Fig 2.3 shows the overall meshing of the plate, whereas fig 2.4 shows the meshing near the crack.

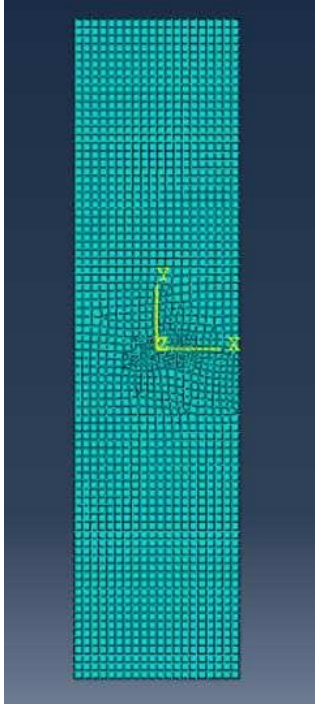


Fig 2.3: The meshed plate

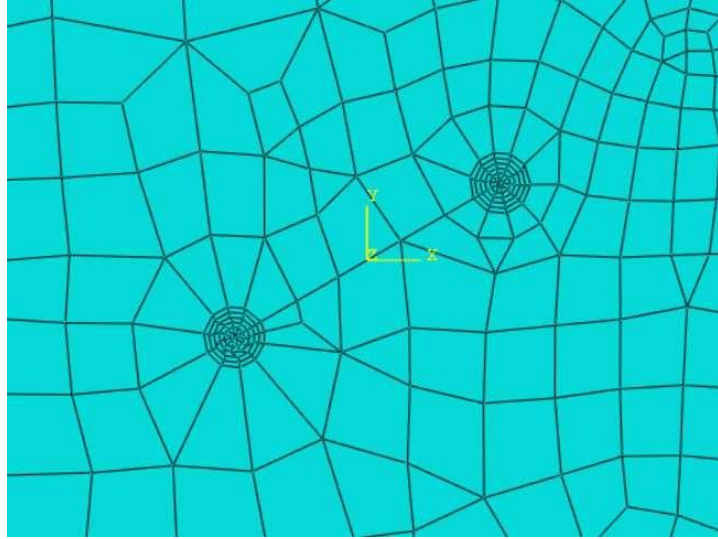


Fig 2.4: Meshing near the crack

Results

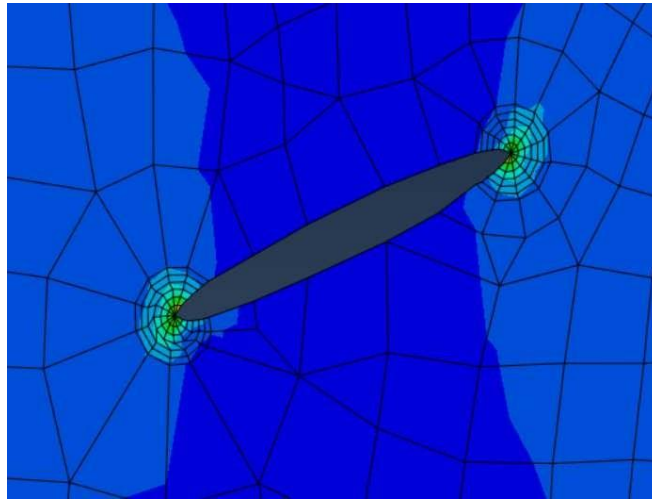


Fig 2.5: Crack opening after loading

(a) Stress Intensity Factor:

Analytically the solution of Mode I and Mode II stress intensity factor for inclined crack case is given by:

$$K_I = \sigma_{\infty} \sqrt{\pi a} \cos^2 \beta$$

$$K_{II} = \sigma_{\infty} \sqrt{\pi a} \sin \beta \cos \beta$$

Hence, $K_I = 132.934 \text{ MPamm}^{\frac{1}{2}}$ and $K_{II} = 76.7495 \text{ MPamm}^{\frac{1}{2}}$.

From Abaqus, $K_I = 135.811 \text{ MPamm}^{\frac{1}{2}}$ and $K_{II} = 78.5187 \text{ MPamm}^{\frac{1}{2}}$.

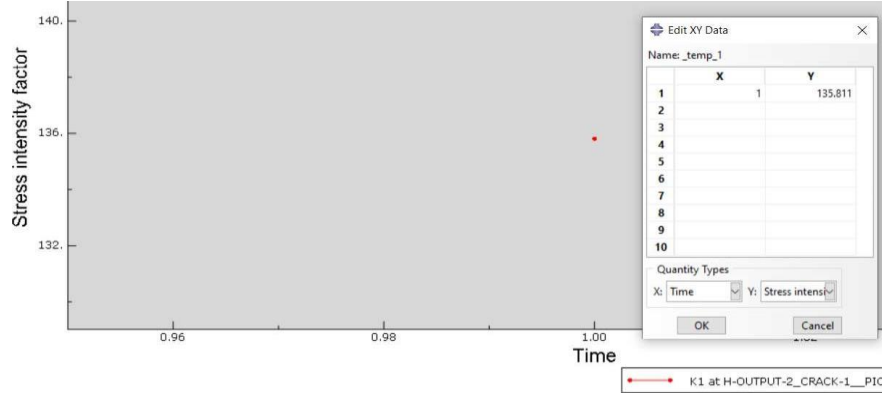


Fig 2.6: Mode I stress intensity factor

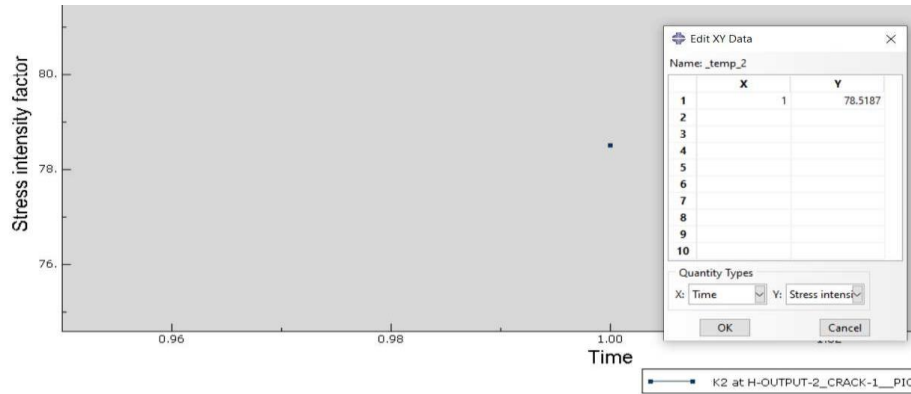


Fig 2.7: Mode II stress intensity factor K_{II}

Crack Propagation Angle

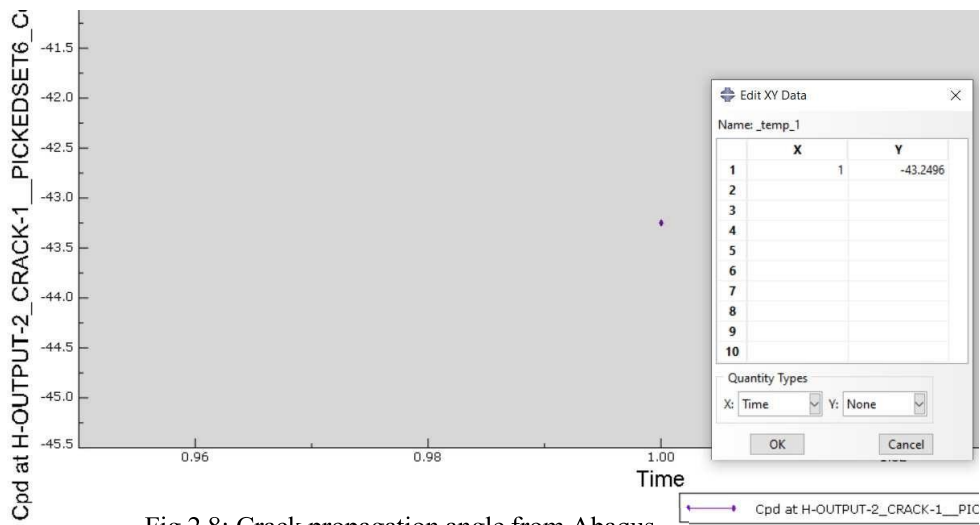


Fig 2.8: Crack propagation angle from Abaqus.

From Abaqus, Crack propagation angle corresponding to maximum tangential/circumferential stress theory is obtained as -43.2496° (See Fig 2.8).

Analytically, the solution is obtained by checking the derivative condition.

$$K_I \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) + K_{II} \left(\cos\left(\frac{\theta}{2}\right) - 2 \sin\left(\frac{\theta}{2}\right)\right) = 0$$

For the mode I and Mode II stress intensity factor obtained analytically, the solution is obtained as -43.201° .

Conclusion

Both the results from Abaqus and analytical part are found to be nearly the same.

Reference:

Fracture Mechanics, C

